

Tut:-

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Ques:- Use Taylor's method of order two to approximate the solⁿ for each of the following initial value problem:
(i) $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, $h = 0.5$

Initial value problem,

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

Discretisation
 $y_0 \rightsquigarrow \dots \rightsquigarrow y_n$
 $t = t_0 \qquad t = t_0 + nh$

Taylor's series,

$$y(t+h) = y(t) + hy'(t) + \frac{h^2}{2!} y''(t) + \frac{h^3}{3!} y'''(t) + \dots$$

$$\text{or } y_{n+1} = y_n + hy'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \dots$$

$$\begin{array}{ccccccc} y_0 & y_1 & y_2 & y_3 & \dots & y_n & y_{n+1} \\ \hline t=t_0 & t=t_1 & t & t_3 & & t_n & t_{n+1} \\ \hline \xleftarrow{h} & \xleftarrow{h} & \xleftarrow{h} & & & \xleftarrow{h} & \end{array}$$

$$\text{So, } y_1 = y_0 + hy'_0 + \frac{h^2}{2!} y''_0 \quad (\text{order } 2 \rightarrow \text{given in ques})$$

$$y' = te^{3t} - 2y$$

$$y'' = 3te^{3t} + e^{3t} - 2y'$$

$$y'' = 3te^{3t} + e^{3t} - 2(te^{3t} - 2y)$$

$$y'' = te^{3t} + e^{3t} + 4y$$

$$y'' = e^{3t}(t+1) + 4y$$

$$\begin{aligned} y''_0 &= e^{3t_0}(t_0+1) + 4y_0 \\ &= e^{3 \times 0}(0+1) + 4(0) \\ y''_0 &= 1 \end{aligned}$$

using initial value,

$$\text{at } t_0 = 0, \quad y_0 = 0$$

$$y'_0 = te^{3t} - 2y_0$$

$$y'_0 = 0e^0 - 2(0) = 0$$

Find the soln for

Verification

$t=10$

$t=0$

in ques")

$+1)+4y_0$
 $+4(0)$

$$n=0 \quad y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0$$

$$y_1 = 0 + (0.5)(6) + \frac{(0.5)^2 \times 1}{2} = \frac{0.25}{2} = 0.125$$

$$n=1 \quad y_2 = y_1 + h y'_1 + \frac{h^2}{2!} y''_1$$

$$y' = te^{3t} - 2y$$

$$y'_1 = ?$$

$$y''_1 = ?$$

$$\cancel{y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0} \quad y'_1 = t_1 e^{3t_1} - 2y_1 \quad (t_1 = 0.5)$$

$$= (0.5)e^{3 \times 0.5} - 2(0.125)$$

$$y'_1 = 1.99084$$

$$y''_1 = e^{3t_1}(t_1 + 1) + 4y_1$$

$$= e^{3 \times 0.5}(0.5 + 1) + 4(0.125)$$

$$y''_1 = 2.222533606$$

$$y_2 = 2.023236$$

② Discretise the following initial value problem

(i) $y' = t+y$; $y(1)=0$

(ii) $y' = -y^2$; $y(1)=1$

Using Euler's method, compute $y(1.2)$ with $h=0.1$ in each case.

$$y_{n+1} = y_n + hf(x_n, y_n)$$

(i) $\frac{dy}{dt} = f(t, y) = t+y$; $y(1)=0$

$t_0 = 1$

$y_0 = 0$ $y_0 = y_1$ y_2

$t_0 = 1$ $t_1 = 1.1$ $t_2 = 1.2$

$y_1 = y_0 + hf(t_0, y_0)$

$y_1 = 0 + (0.1)f(1, 0)$

$y_1 = 0 + (0.1)(1+0)$

$y_1 = 0.1$

$y_2 = y_1 + hf(t_1, y_1)$

$= 0.1 + hf(1.1, 0.1)$

$= 0.1 + (0.1)(1.1 + 0.1)$

$= 0.1 + (0.1)(1.2)$

$= (0.1)(2.2)$

$y_2 = 0.22$

$\frac{dy}{dt} = t+y \Rightarrow \frac{dy}{dt} - y = t$

If $\int e^{-t} dt = -e^{-t}$

$y e^{-t} = \int e^{-t} \cdot t dt$

$y e^{-t} = -\frac{t}{2} e^{-t} + C$

$y = -\frac{t}{2} + C e^t$

$y e^{-t} = -\frac{t}{2} e^{-t} + C$

$y = -\frac{t}{2} + C e^t$

$t=1, y=0$

$0 = -\frac{1}{2} + C e$

$2 = C e$

$C = \frac{2}{e}$

$y = -\frac{t}{2} + \frac{2}{e} e^t$

at $t=1.2$

$y = 0.2428$

(ii) $t_0 = 1$

$y_0 = 1$

$y_1 = y_0 + hf(t_0, y_0)$

$= 1 + (0.1)f(1, 1)$

$= 1 + (0.1)(1-1)$

$= 1 + 0 = 1$

$y_1 = 1$

$y_0 = 1$ y_1 y_2

t_0 t_1 t_2

1 1.1 1.2

t_0 t_1 t_2

Ans = 0.0142314

$$y_2 = 0.9 + (0.1)(-0.9^2) = 0.819$$

$$y_2 = y_1 + hf(t_1, y_1) = 1.2 + (0.1)(1.1 + 1.2) = 1.2 + (0.1)(2.3) = 1.2 + 0.23 = 1.43$$

$$y_2 = 1.43$$

$$\frac{dy}{dt} = -y^2 \Rightarrow \int \frac{dy}{y^2} = \int dt \quad \left. \frac{1}{y} \right|_1^{1.2} = t \Big|_0^{1.2}$$

$$\frac{1}{y} - 1 = 1.2 - 1 \Rightarrow \frac{1}{y} - 1 = 0.2 \Rightarrow \frac{1}{y} = 1.2 \Rightarrow y = 0.833$$

$$y = 0.833$$

Estimate $\int_0^{0.5} \int_0^{0.5} \frac{\sin xy}{1+xy} dx dy$, using trapezoidal & Simpson's rule. $h=k=0.25$

Trapezoidal:- $\int_a^b f(x) dx = \left(\frac{b-a}{2}\right) (f(a) + f(b))$

$$\int_0^{0.5} \int_0^{0.5} \frac{\sin xy}{1+xy} dx dy$$

$$(0.25) \int_0^{0.5} \left(\left(\frac{0.25}{2}\right) \left\{ f(0) + f(0.5) + 2f(0.25) \right\} \right) dy$$

$$(0.25) \int_0^{0.5} \left(\frac{\sin 0}{1+0} + \frac{\sin(0.5y)}{1+0.5y} + \frac{2 \sin(0.25y)}{1+0.25y} \right) dy$$

$$(0.125) \left[\left(\frac{0.25}{2}\right) \left\{ \frac{\sin(0.5 \times 0)}{1+0.5 \times 0} + \frac{2 \sin(0.5 \times 0.25)}{1+0.5 \times 0.25} + \frac{\sin(0.5 \times 0.5)}{1+0.5 \times 0.5} \right. \right.$$

$$\left. + \frac{2 \sin 0}{1+0} + \frac{4 \sin(0.25^2)}{1+0.25^2} + \frac{2 \sin(0.25 \times 0.5)}{1+0.25 \times 0.5} \right\}$$

$$= 0.01369311$$

$$Ans = 0.01369311$$

Simpson: $\frac{h}{3} \{ f(x_0) + 4 \text{ odd} + 2 \text{ even} + f(x_n) \}$

$= 0.0140629$

④ Evaluate: $h = \kappa = 0.25$ (Trapezoidal)

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$$\int_0^1 \int_1^2 \frac{2xy}{(1+x^2)(1+y^2)} dy dx$$

$$\int_0^1 \left\{ \left(\frac{0.25}{2} \right) \left[\frac{2x}{(1+x^2)^2} + \frac{4x}{5(1+x^2)} + 2 \left(\frac{2x \times 1.25}{(1+x^2)(1+1.25^2)} + \frac{2x \times 1.5}{(1+x^2)(1+1.5^2)} + \frac{2x \times 1.75}{(1+x^2)(1+1.75^2)} \right) \right] \right\} dx$$

$$\left(\frac{0.25}{2} \right) \int_0^1 \left[\frac{x}{(1+x^2)} + \frac{4x}{5(1+x^2)} + \frac{86x}{41(1+x^2)} + \frac{12x}{13(1+x^2)} + \frac{2x}{1+x^2} \right] dx$$

$$\left(\frac{0.25}{2} \right) \left(\frac{5.53583}{9.56435} \right) \int_0^1 \frac{x}{1+x^2} dx$$

$$\left(\frac{0.25}{2} \right) \left[\frac{0}{1+0} + \frac{1}{1+1} + 2 \left[\frac{0.25}{1+0.25^2} + \frac{0.5}{1+0.5^2} + \frac{0.75}{1+0.75^2} \right] \right]$$

$$\left(\frac{0.25}{2} \right) \left(\frac{9.56435}{5.53583} \right) (0.2019)$$

$= 0.1397186506$

Runge

$$\frac{dy}{dx} =$$

$$y_{n+1} =$$

where,

K

K₂

f

Ques Find y(2) using Runge

$$\frac{dy}{dx} = x$$

for n=0

$$y_1 =$$

$$K_1 = f(x_0,$$

$$K_2 = f(x_0,$$

$$=$$

$$K_3 = f(x_0,$$

$$K_4 = f(x_0,$$

Runge Kutta's Method of fourth Order:-

$$\frac{dy}{dx} = f(x, y) ; y(x_0) = y_0 \text{ (IVP)}$$

$$y_{n+1} = y_n + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4), \quad n=0, 1, 2, \dots \rightarrow (i)$$

where,

$$K_1 = f(x_n, y_n)$$

$$K_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{K_1}{2}\right)$$

$$K_3 = f\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right)$$

$$K_4 = f(x_n + h, y_n + K_3)$$

Ques Find $y(2.2)$ for IVP $y' = x(y-x)$, $y(2) = 3$ in $[2, 2.4]$ using Runge-Kutta's method with step-size $h=0.2$.

$$\frac{dy}{dx} = \frac{x(y-x)}{f(x, y)}$$

$$y(2) = 3$$

$$x_0 = 2, y_0 = 3$$

for $n=0$ in eqⁿ (i),

$$y_1 = y_0 + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

$$K_1 = f(x_0, y_0) = x_0(y_0 - x_0) = 2(3-2) = 2$$

$$K_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right) = f\left(2 + \frac{0.2}{2}, 3 + \frac{2}{2}\right) = f(2.1, 4)$$

$$= 2.1(4 - 2.1) = 2.1 \times 1.9 = 3.99$$

$$K_3 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) = f\left(2.1, 3 + \frac{3.99}{2}\right) = f(2.1, 4.995)$$

$$= 6.0795$$

$$K_4 = f(x_0 + h, y_0 + K_3) = f(2.2, 3 + 6.0795) = f(2.2, 9.0795)$$

$$= 15.1349 \approx 14.65695$$

$$y_1 = y_0 + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

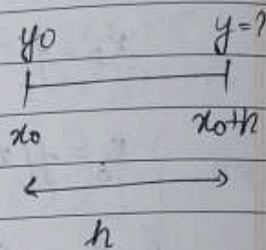
$$= 3 + 0.2 \left(2 + 2(3.99 + 6.0795) + 14.65695 \right)$$

$$= 11.226531667 \quad 4.24246$$

Taylor series method :-

$$y(x_0 + h) = y(x_0) + h y'(x_0) + \frac{h^2}{2} y''(x_0) + \dots$$

$$y' = \frac{dy}{dx} = f(x, y) \quad ; \quad y(x_0) = y_0$$



$$y(x_1) = y(x_0) + h y'(x_0) + \frac{h^2}{2!} y''(x_0)$$

Ques. Obtain the approximated value of $y(1.3)$ for the IVP:
 $y' = -2xy^2$, $y(1) = 1$, by using Runge-Kutta's method of second order
 step size $h = 0.1$

Also compare it with exact solⁿ.

$$y(x_0 + h) = y(x_0) + h y'(x_0) + \frac{h^2}{2!} y''(x_0)$$

$$y(1+0.1) = y(1) + (0.1) y'(1) + \frac{(0.1)^2}{2!} y''(1)$$

$$y(1.1) = 1 + (0.1)(-2 \times x_0 \times y_0^2) + \frac{(0.1)^2}{2!} \left\{ \frac{d}{dx} (-2xy^2) \right\}$$

$$y(1.1) = 1 + (0.1)(-2 \times 1 \times 1^2) + \frac{(0.1)^2}{2!} (-2(xy y' + y^2)) \Big|_{x_0, y_0}$$

$$= 1 - 0.2 + \frac{0.01}{2} (-2(1 \times 1 \times (-2) + 1^2))$$

$$= 1 - 0.2 - 0.003$$

$$= 0.797$$

$$y(1.1) = \frac{1 - 0.2 + 0.01x - 2(2 \times 1 \times 1 \times -2 + 1)}{2}$$

$$= 1 - 0.2 - 0.01x - 3 = 1 - 0.2 + 0.03 = 1.03 - 0.2$$

$$y(x_0 + h) = y(x_0) + hy'(x_0) + \frac{h^2}{2!} y''(x_0) = 0.83 //$$

$$\Rightarrow y(1.2) = y(1.1 + 0.1)$$

$$= y(1.1) + (0.1)y'(1.1) + \frac{(0.1)^2}{2!} y''(1.1)$$

$$= 0.83 + (0.1)(-1.51558) + \frac{(0.1)^2}{2!} (4.157)$$

$$= 0.699227$$

$$y' = -2xy^2$$

$$= -2 \times 1.1 \times 0.83^2$$

$$= -1.51558$$

$$y'' = -2(2xyy' + y^2)$$

$$= -2(2 \times 1.1 \times 0.83 \times -1.51558 + 0.83^2)$$

$$= 4.157$$

$$\Rightarrow y(1.3) = y(1.2 + 0.1) = y(1.2) + (0.1)y'(1.2) + \frac{(0.1)^2}{2!} y''(1.2)$$

$$y(1.2) = 0.699$$

$$y'(1.2) = -2 \times 1.2 \times 0.699^2$$

$$= -1.1126$$

$$\Rightarrow 0.5965255$$

$$y'' = -2(2 \times 1.2 \times 0.699 \times -1.1126 + 0.699^2)$$

$$= 2.9571$$

Exact:-

$$y' = -2xy^2$$

$$\frac{dy}{dx} = -2xy^2$$

$$y \int \frac{dy}{y^2} = -2x \int dx$$

$$+\frac{1}{y} \Big|_1^y = \frac{-2x^2}{2} \Big|_1^x$$

$$\frac{1}{y} - 1 = x^2 - 1$$

$$\boxed{y = \frac{1}{x^2}}$$

⇒ n th order Ordinary differential equation can be transformed into system of n first order differential equation

eg $y'' + \alpha y' + \beta y = r \rightarrow (i)$
 consider, $y = y_1$ (new variable)
 $y' = y_2$ (new variable)

then,

$$y_1' = y_2 \Rightarrow y_1' = y_2 \rightarrow (ii)$$

Also, $y_2' = y'' = r - \alpha y_1 - \beta y_2$ (using (i))

$$y_2' = r - \alpha y_1 - \beta y_2 \rightarrow (iii)$$

Thus we have, $\left. \begin{array}{l} y_1' = y_2 \\ y_2' = r - \beta y_1 - \alpha y_2 \end{array} \right\}$ system of two first order differential equation.

or $\vec{y}' = \vec{f}$ where, $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ & $\vec{f} = \begin{bmatrix} y_2 \\ r - \beta y_1 - \alpha y_2 \end{bmatrix}$

$$\vec{y}' = \vec{f}(\vec{x}, \vec{y}), \quad \vec{y}(x_0) = \vec{y}_0$$

a) Euler's formula: $\vec{y}_{n+1} = \vec{y}_n + h \vec{f}(x_n, \vec{y}_n); n=0, 1, 2, \dots$

b) Backward euler's formula: $\vec{y}_{n+1} = \vec{y}_n + h \vec{f}(x_{n+1}, \vec{y}_{n+1})$
 $; n=0, 1, 2, \dots$

c) RK method of 4th order:

$$\vec{y}_{n+1} = \vec{y}_n + \frac{h}{6} (\vec{a} + 2\vec{b} + 2\vec{c} + \vec{d}) \quad n=0, 1, 2, \dots$$

where

$$\begin{aligned}\vec{a} &= \vec{f}(x_n, \vec{y}_n) \\ \vec{b} &= \vec{f}(x_n + h/2, \vec{y}_n + \vec{a}/2) \\ \vec{c} &= \vec{f}(x_n + h/2, \vec{y}_n + \vec{b}/2) \\ \vec{d} &= \vec{f}(x_n + h, \vec{y}_n + \vec{c})\end{aligned}$$

Ques Solve IVP for a damped mass spring: $y'' + 2y' + 0.75y = 0$,
 $y(0) = 3$ & $y'(0) = -2.5$ by Euler's method for system
 with $h = 0.2$ for $0 \leq x \leq 1$ (x is time here)

find
 $y(0.2)$

$$x_0 = 0 \quad x_1 = 0.2 \quad x_2 = 0.4 \quad x_3 = 0.6 \quad x_4 = 0.8 \quad x_5 = 1$$

Consider $y = y_1$ & $y' = y_2$

$$y' = y_1' = y_2 \Rightarrow y_2 = y_1' \quad (i)$$

$$y'' + 2y' + 0.75y = 0$$

$$y_1' = y_2, \quad y_2' = y'' = -2y' - 0.75y \quad (ii)$$

2 first order
 differential
 equation

$$\begin{cases} y_1' = y_2 \\ y_2' = -2y_1' - 0.75y_1 \end{cases}$$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} y_2 \\ -2y_1' - 0.75y_1 \end{bmatrix}$$

$$\vec{y}' = f(x, \vec{y}) \quad , \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Initial condⁿ: $y(0) = 3$
 $y_1(0) = 3$
 $y_1' = 3$

$$\begin{aligned}y_1'(0) &= -2.5 \\ y_2(0) &= -2.5 \\ y_2, 0 &= -2.5\end{aligned}$$

$$\vec{y}_0 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Euler's method

for $n = 0$,

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix}$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix}$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix} = \begin{bmatrix} 3 \\ -2.5 \end{bmatrix}$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix}$$

since y_1
 y_2

Solve the IVP
 Using R-K

$\Rightarrow$ let $y = y_1$

y_1'

y_2'

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ -2y_1' - 0.75y_1 \end{bmatrix}$$

$$\vec{y}_0 = \begin{bmatrix} y_{1,0} \\ y_{2,0} \end{bmatrix} = \begin{bmatrix} 3 \\ -2.5 \end{bmatrix}$$

Euler's method, $\vec{y}_{n+1} = \vec{y}_n + f(x_n, \vec{y}_n) h$

$$\begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix} + h \vec{f}\left(x_n, \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix}\right)$$

for $n=0$,

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix} = \begin{bmatrix} y_{1,0} \\ y_{2,0} \end{bmatrix} + h \vec{f}(x_0, \begin{bmatrix} y_{1,0} \\ y_{2,0} \end{bmatrix})$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix} = \begin{bmatrix} y_{1,0} \\ y_{2,0} \end{bmatrix} + h \begin{bmatrix} y_{2,0} \\ -0.75y_{1,0} - 2y_{2,0} \end{bmatrix}$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix} = \begin{bmatrix} 3 \\ -2.5 \end{bmatrix} + (0.2) \begin{bmatrix} -2.5 \\ -0.75 \times 3 - 2 \times -2.5 \end{bmatrix} \quad \left(\vec{f} = \begin{bmatrix} y_2 \\ -0.75y_1 - 2y_2 \end{bmatrix} \right)$$

$$\begin{bmatrix} y_{1,1} \\ y_{2,1} \end{bmatrix} = \begin{bmatrix} 2.5 \\ -1.95 \end{bmatrix}$$

since $y_1 = y$
 $y_{1,1} = y(0.2) = 2.5$

Ex. Solve the IVP $y'' + 4y = \cos x$, $y(0)=1$, $y'(0)=1$
 Using R-K method $h=0.2$ on $(0, 0.4)$

$\Rightarrow$ Let $y = y_1$ & $y' = y_2$

0	0.2	0.4
x_0	x_2	x_4

$y_1' = y_2$
 $y_2' = y'' = \cos x - 4y$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} y_2 \\ \cos x - 4y \end{bmatrix}$$

$$\text{ans} = \vec{y}(0.2) = \begin{bmatrix} 0.940733 \\ -0.581517 \end{bmatrix}$$

$$y_{1,0} = 1 \quad \vec{y}' = \vec{f}$$

$$\& y_{2,0} = 1$$

$$\vec{y}(0) = \vec{y}_0$$

$$\text{where } \vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \& \vec{f} = \begin{bmatrix} y_2 \\ \cos x - y_1 \end{bmatrix} \quad \& \vec{y}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{y}_{n+1} = \vec{y}_n + h \frac{[\vec{a} + 2\vec{b} + 2\vec{c} + \vec{d}]}{2}$$

$$n=0, \begin{bmatrix} y_{1,n+1} \\ y_{2,n+1} \end{bmatrix} = \begin{bmatrix} y_{1,n} \\ y_{2,n} \end{bmatrix} + h_2 [\vec{a} + \vec{a} + 2\vec{c} + \vec{d}]$$